Lab 7

Buffer: Assemble, Troubleshoot, and Evaluate Performance

Shaymon Madrid

Samuel Bourgeous

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ECEN 299

Objectives

The objectives of this lab were to assemble the buffer board, verify that it was assembled correctly, and then test the board like we tested previous boards to verify it is functional as required.

Procedure

Equipment and supplies

- Soldering Iron and Solder
- Microscope
- PCB Components
- Laptop
- Speakers
- Function Generator
- Oscilloscope

Procedure

For this lab, we began by reviewing some best practices and trainings for soldering. We also verified that equipment items such as the soldering iron and the microscope were working. To assemble the daughterboard, first, the op amp was soldered to the PCB; taking care to make sure there was no sign of bridging between each pad or pin on the chip and that the board was straight on the PCB. After this, the resistors were then soldered on to their respective pads as required from previous design requirements and arrangement. In referencing these previous labs and schematics we made sure that each resistor was properly placed on the board for the circuit to be properly complete. The header pins were then soldered on then the board was cleaned. After this the board was checked with Brother Smith with his tester and passed his check.

Once the board was completed and its assembly verified as correct, the board was inserted into the line buffer slot of the chassis and powered. We also placed the jumpers as shown in the lab instructions to ensure we were passing our signals through the buffer board for analysis. To test this we used the left buffer in pin with signals from a function generator at 2Vpp (see table 1 for values) then repeated the analysis on the right channel.

Vin=2Vpp	Frequency	Vin(pp)	Vout(pp)	Gain(A=Vout/Vin)
Right Channel	20Hz	2.12V	6.08V	2.87
	200Hz	2.08V	6.16V	2.96
	2kHz	2.08V	6.0V	2.88
	10kHz	2.04V	6.0V	2.94
Left Channel	20Hz	2.04V	6.16V	3.02
	200Hz	2.08V	6.0V	2.88
	2kHz	2.04V	6.0V	2.94
	10kHz	2.08V	6.0V	2.88

Table 1: Frequency Response

This analysis was done by adjusting the frequency on the function generator for the frequencies found in the table above while using an oscilloscope to measure the output value as seen (for the left channel) in figures 1 and 2.

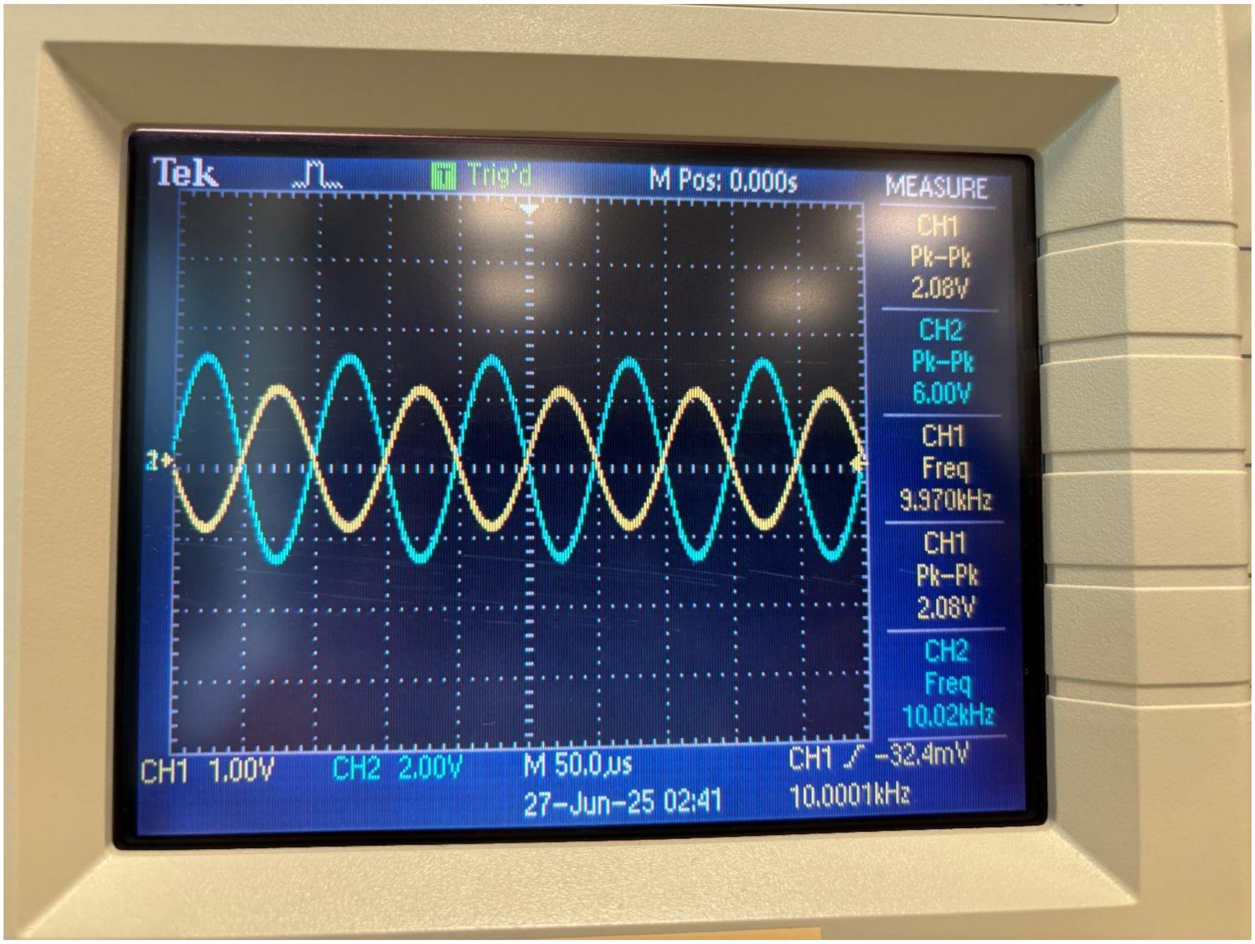

Figure 1: 10kHz

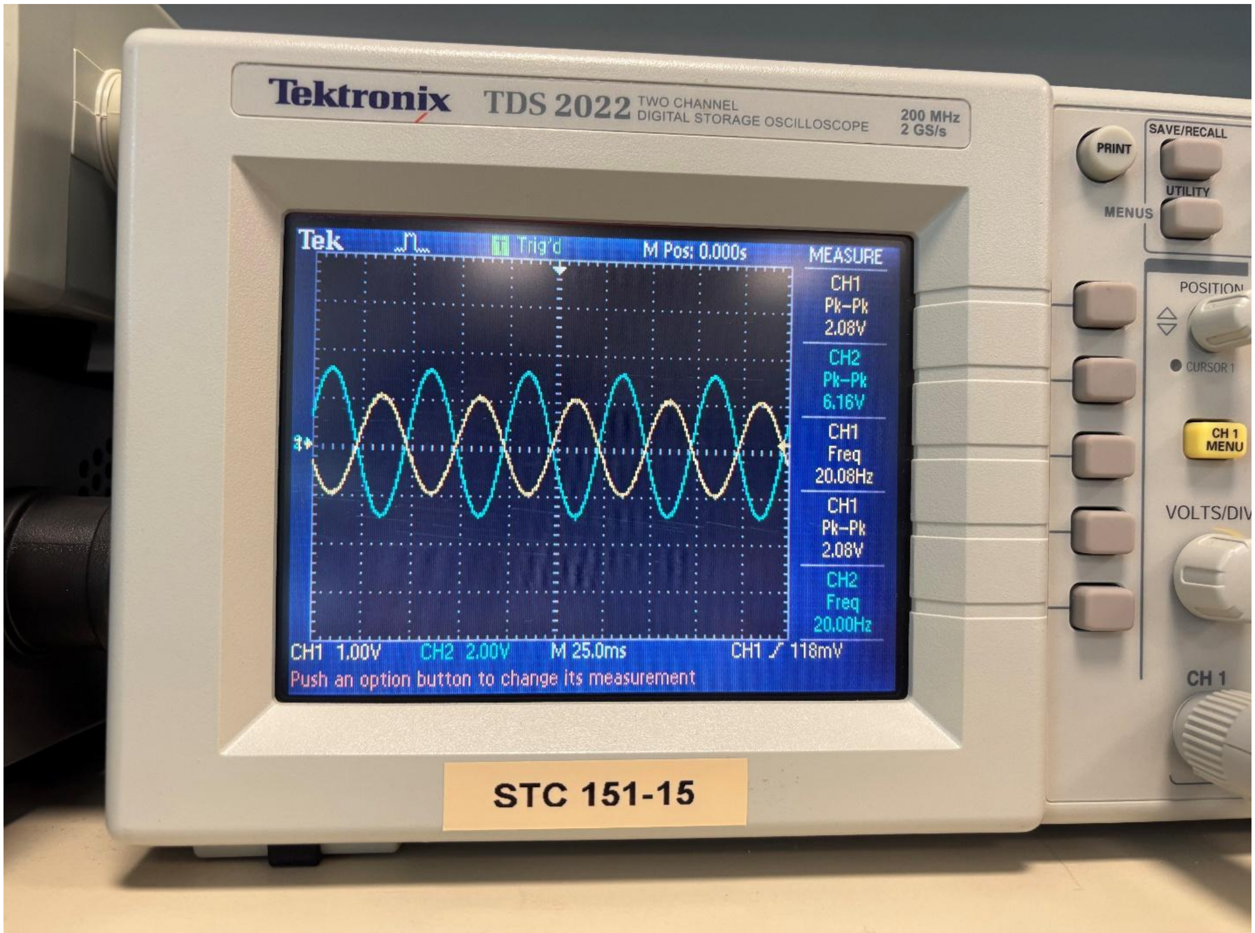

Figure 2: 20kHz

From the information gathered, this board meets our requirements and should function as required. The maximum deviation for this board is 0.2 dB.

Below are figures 3, 4, and 5 to show the lab station after this session.

Figure 3

Figure 4

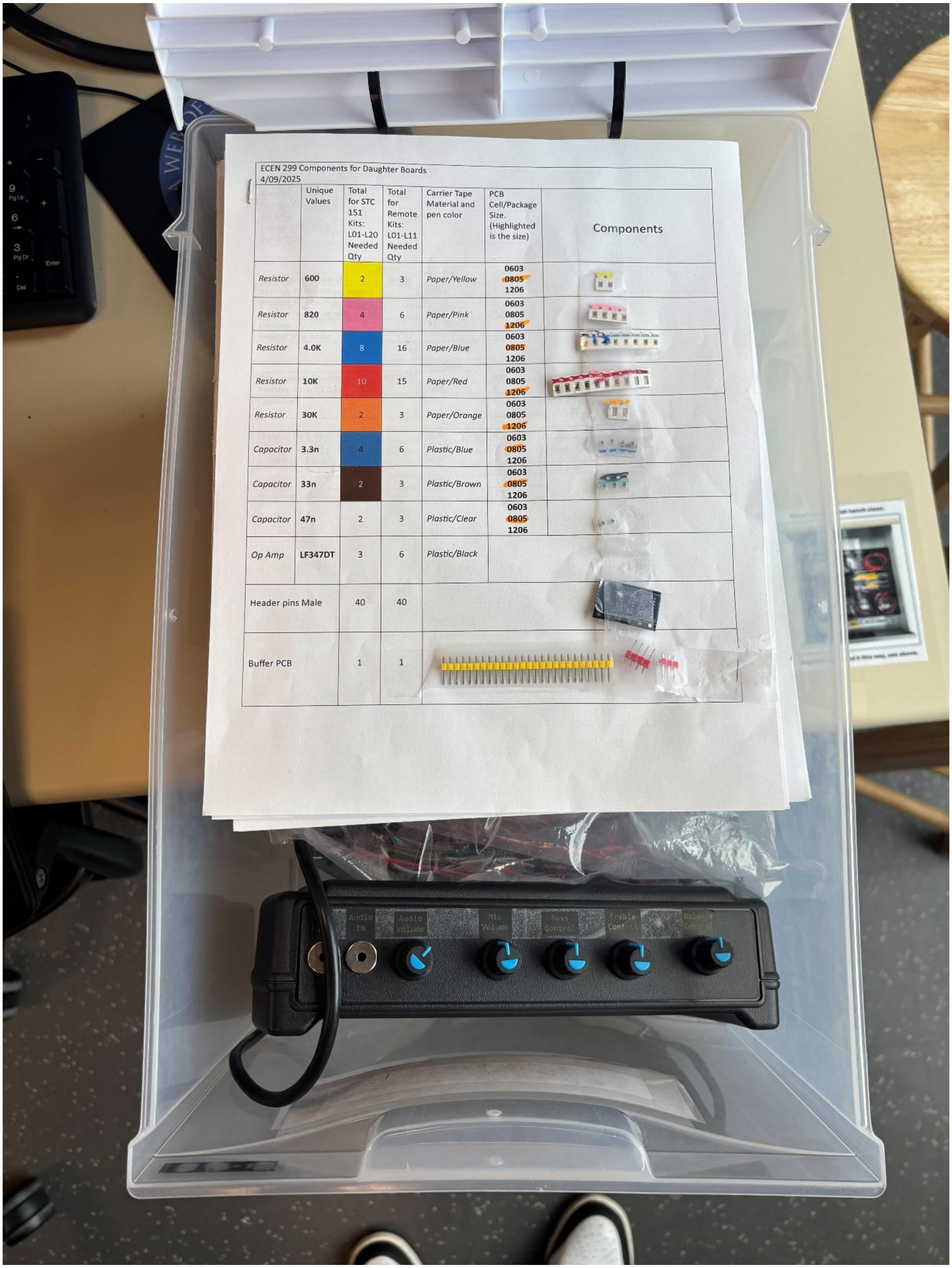
ECEN 299 Components for Daughter Boards
4/09/2025
Unique Values
Total for STC 151 Kits: L01-L20 Needed Qty
Total for Remote Kits: L01-L11 Needed Qty
Carrier Tape Material and pen color
PCB Cell/Package Size. (Highlighted is the size)
Components
Resistor 600 2 3 Paper/Yellow 0603 0805 1206
Resistor 820 4 6 Paper/Pink 0603 0805 1206
Resistor 4.0K 8 16 Paper/Blue 0603 0805 1206
Resistor 10K 10 15 Paper/Red 0603 0805 1206
Resistor 30K 2 3 Paper/Orange 0603 0805 1206
Capacitor 3.3n 4 6 Plastic/Blue 0603 0805 1206
Capacitor 33n 2 3 Plastic/Brown 0603 0805 1206
Capacitor 47n 2 3 Plastic/Clear 0603 0805 1206
Op Amp LF347DT 3 6 Plastic/Black
Header pins Male 40 40
Buffer PCB 1 1
Audio IN
Audio Volume
Mic Volume
Bass Control
Treble Control

Figure 5

Conclusion

Part of this lab session also included doing some research on potentiometers with specific references to the words terminal, wiper, taper, dual-gang pot, linear taper and logarithmic taper. In regards to potentiometers, the terminal is one of the connectors on the potentiometer, most have three of these. The wiper is the contact inside which can be moved by the user allowing the resistance to be adjusted between the wiper and each terminal end. The taper is how the resistance changes as the shaft is rotated and is the relationship between the angle and output voltage such as linear and logarithmic relationships. Linear taper potentiometers increase resistance in a linear relationship relative to the shaft, whereas the logarithmic taper potentiometers change resistance exponentially. Dual-Gang potentiometers have two potentiometers in one package adjusted by just one shaft. Each controls a separate circuit allowing simultaneous adjustment.

Logarithmic potentiometers are good for volume as human hearing itself is logarithmic making the volume control more even and natural sounding. Linear potentiometers are good for balance control as it allows us to more equally and predicably change the balance between each speaker.

This lab was passed off on Saturday by Ryan, one of the TA’s for this class. Some of the challenges I encountered in this lab included the soldering portion; it was difficult at times to ensure correct contact and that the components were straight. Other issues included bridging which needed to be resolved. Though different, this lab wasn’t difficult just time consuming on assembling the board with things being so small and hard at times to keep where I needed them to be while soldering.